Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

CSE4204 — Mobile Computing Lab · Final Project Presentation

RESUMIND

AI-Powered Resume Analyzer and Career Copilot

Presented by — Team CSE4204-8A-T05 (Section 8A)

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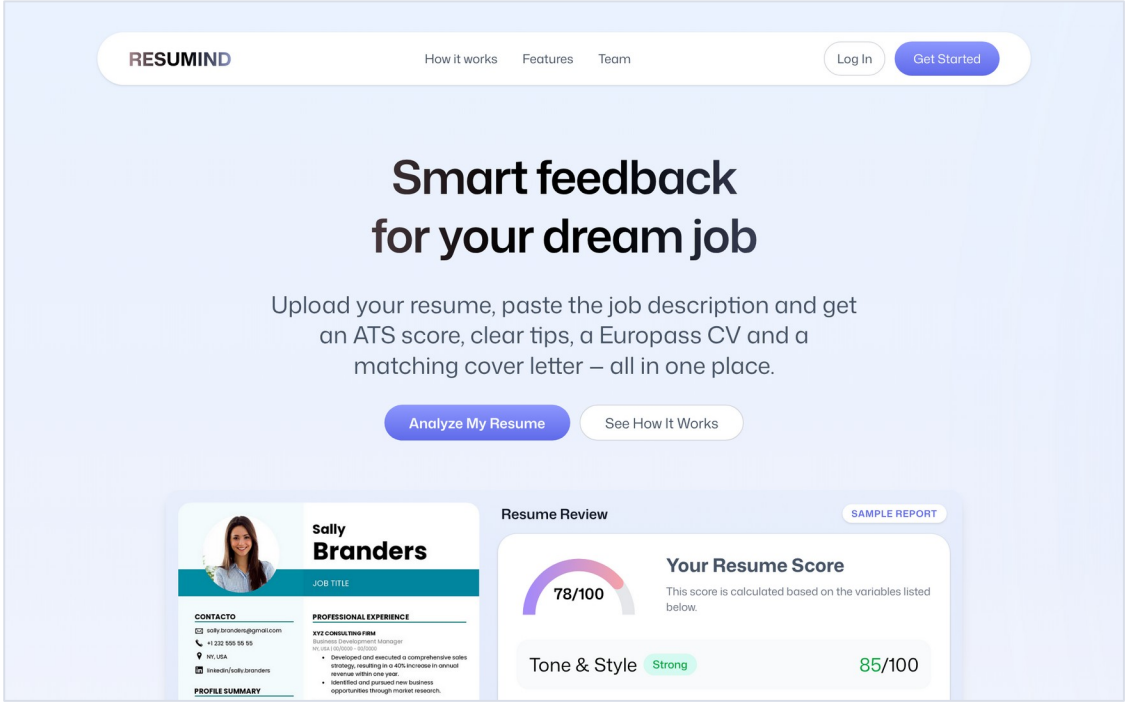
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Project Introduction

RESUMIND is a responsive web application that checks a resume against a real job and helps the candidate improve it.

- Upload a PDF resume and paste the job description
- Get an AI score: overall + ATS, tone, content, structure, skills
- Rewrite the resume as a European-standard (Europass) CV
- Measure the improvement with a before / after score
- Generate a one-page cover letter for the same job
- Works on phone, tablet and desktop; live on the web



Problem Statement

Filtered by ATS

Applicant Tracking Systems reject resumes with missing keywords, weak verbs or unclear structure.

No feedback

Applicants get repeated rejections without knowing what to fix.

Unknown standard

International employers expect a Europass CV that most graduates were never taught.

Time-consuming

Writing a personalised cover letter for every job takes hours.

Existing tools are not enough: Jobscan and Resume Worded are paid and match keywords without context; Reactive Resume is only a builder. None combine analysis, a Europass rewrite and a cover letter for the same job.

Objectives

O1	Secure accounts with bcrypt, JWT and Google Sign-In
O2	Store resumes and analysis history in MongoDB
O3	AI ATS analysis: overall score + 5 categories with tips
O4	Europass rewrite that fixes every tip without inventing facts
O5	Measure the improvement with a before / after score
O6	One-page cover letter tied to the same job
O7	Responsive, mobile-first interface
O8	48 automated tests, CI and public deployment

Key Features

1. Upload

PDF + job title and description

2. Analyse

Overall, ATS, tone, content, structure, skills

3. Europass CV

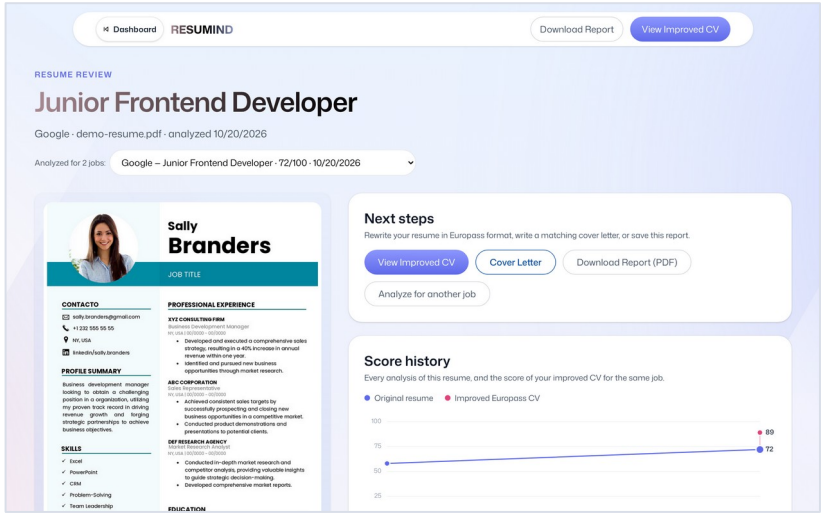
Rewrite, inline edit, change list, PDF

4. Before / after

Same job, new score, history chart

5. Cover letter

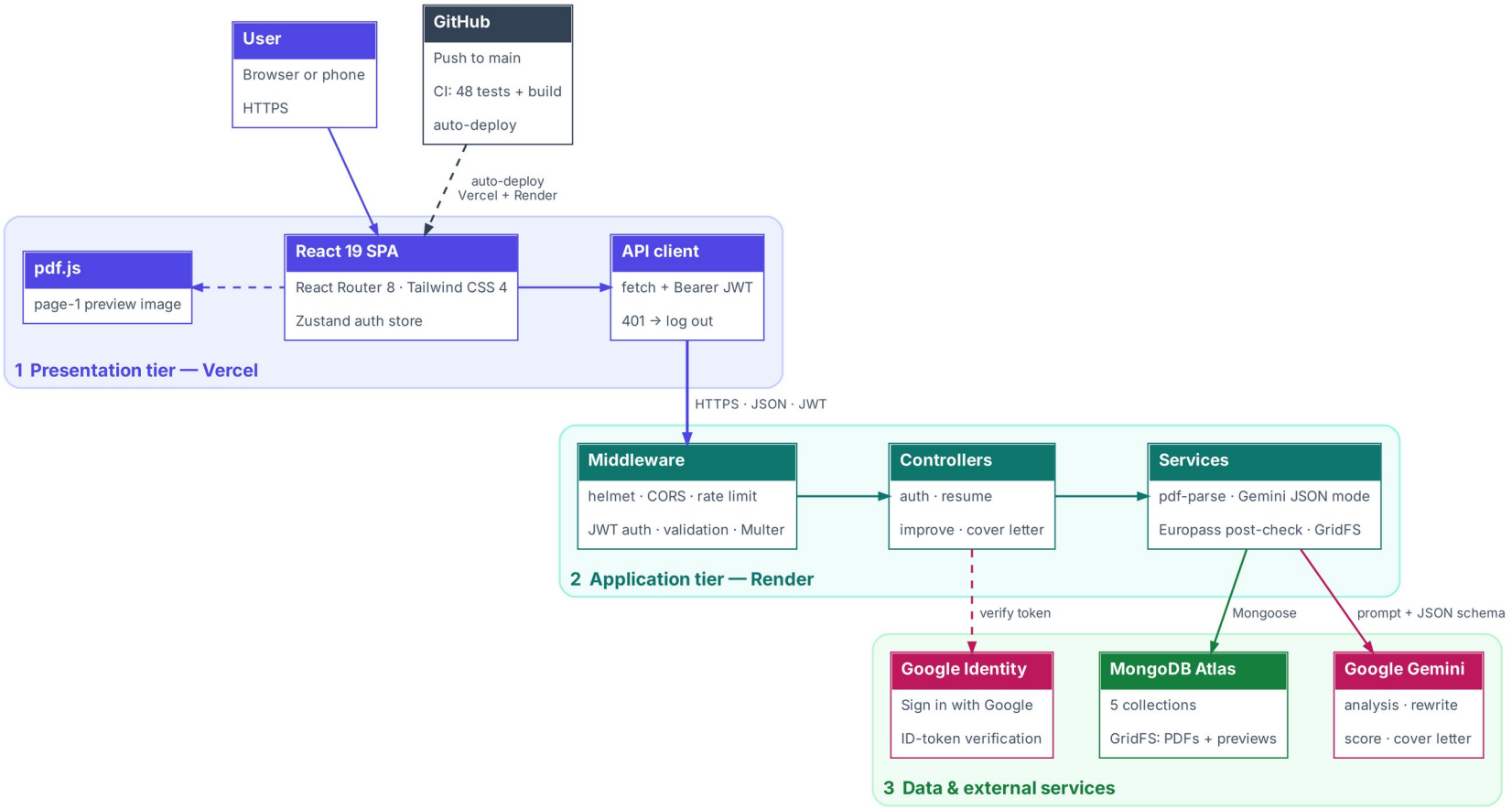
Formal or modern, editable, PDF



Also included

- History dashboard and printable report
- Email/password and Google Sign-In
- Change password, delete account
- Phone, tablet and desktop layouts

System Architecture



Presentation tier
React SPA on Vercel

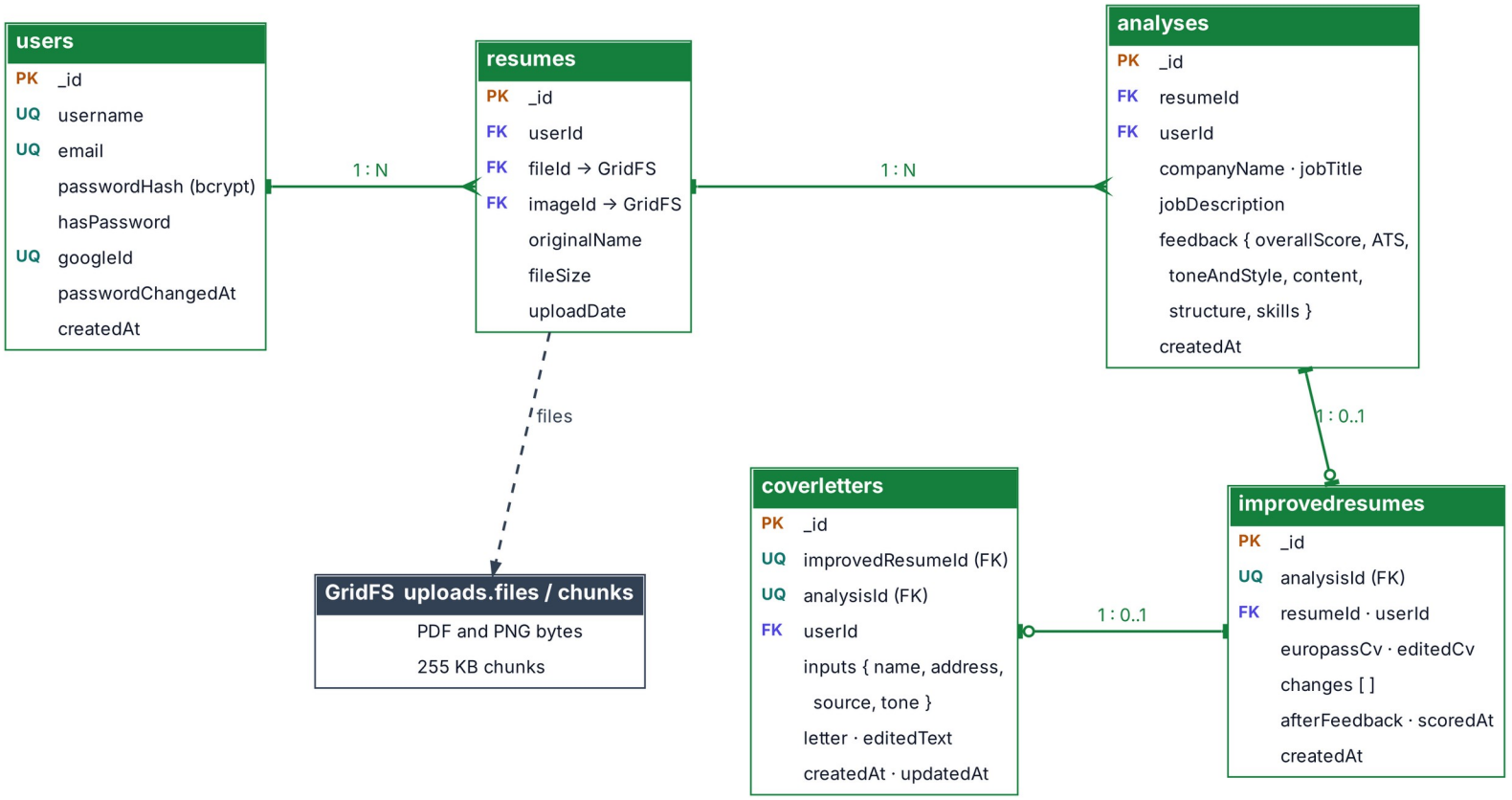
Application tier
Express 5 REST API on Render

Data tier
MongoDB Atlas + GridFS; Google Gemini and Google Identity

Technology Stack

Layer	Technology	Why we chose it
Frontend	React 19, React Router 8, TypeScript, Tailwind CSS 4, Zustand, pdf.js	Component model, type safety, fast mobile-first styling
Backend	Node.js, Express 5, Multer, pdf-parse, express-validator, helmet	Same language as frontend, mature security middleware
Database	MongoDB Atlas, Mongoose 9, GridFS	Nested AI feedback and CVs fit documents; files stored with data
AI	Google Gemini (@google/genai, JSON mode)	Structured output with a JSON schema; free tier
Auth	bcryptjs, jsonwebtoken, google-auth-library	Standard password hashing, stateless tokens, verified Google login
Testing / CI	Vitest, Supertest, mongodb-memory-server, GitHub Actions	Real HTTP tests on every push
Hosting	Vercel (frontend), Render (backend), Docker Compose (local)	Free tiers with automatic deploys

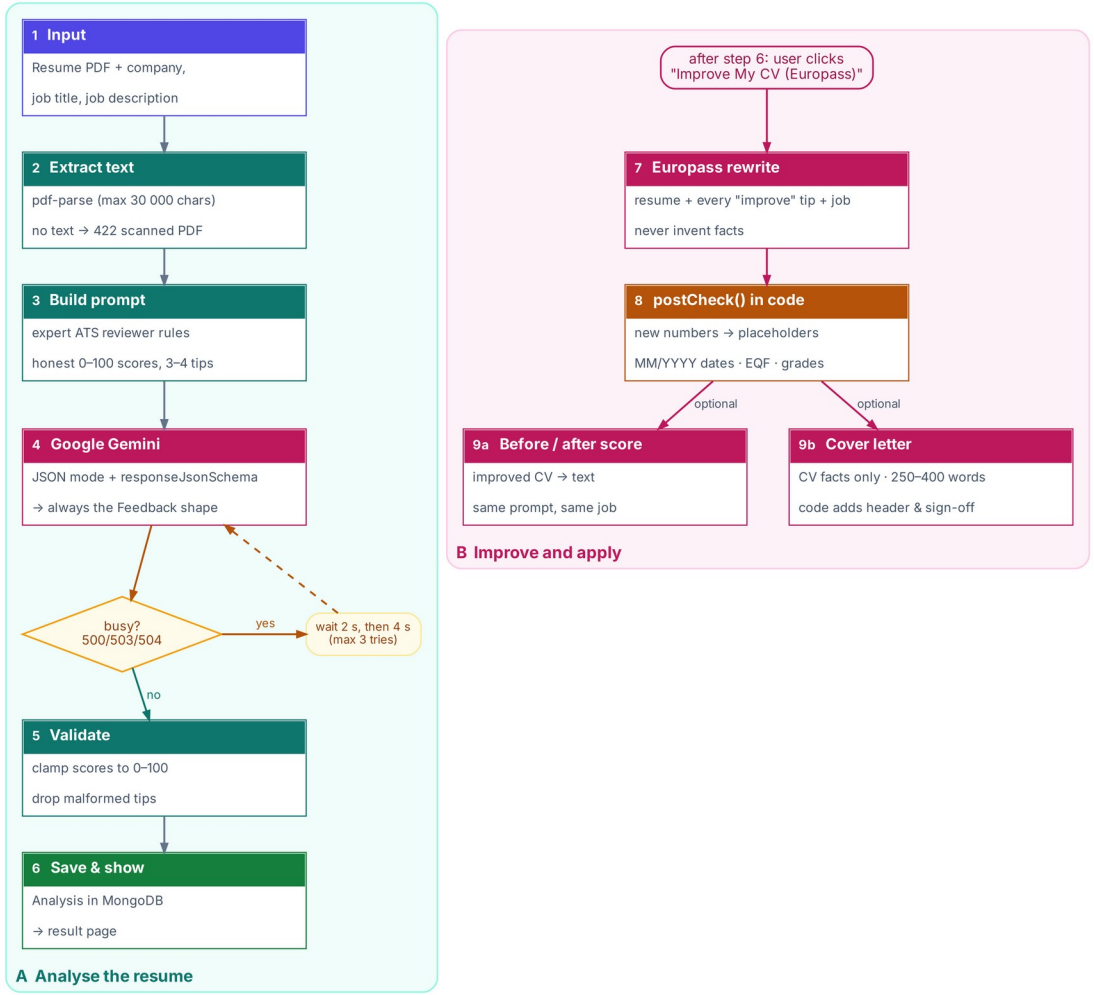
Database Design



users 1–N resumes 1–N analyses 1–0..1 improvedresumes 1–0..1 coverletters

- userId on every document — always filtered by owner
- Unique indexes: one CV per analysis, one letter per CV
- Cascade delete for resume and account
- AI version and user edits stored side by side
- PDFs in GridFS — safe across redeploys

AI Workflow



How the AI is used

- Gemini in JSON mode with a strict schema — the answer always has our exact structure
- Busy errors (500/503/504) are retried after 2 s and 4 s
- Scores are clamped to 0–100 and malformed tips are dropped
- The same service powers analysis, Europass rewrite, before/after score and cover letter
- 15 AI requests per user per hour; the API key stays on the server

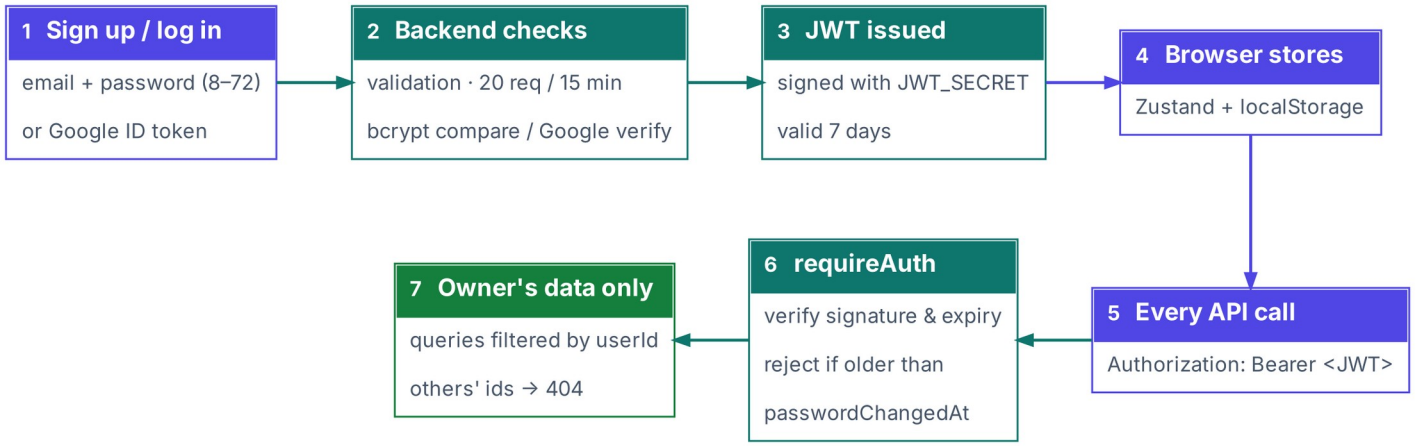
Keeping the AI Honest

After Gemini answers, postCheck() in our code fixes what the model gets wrong:

AI wrote	After post-check
improved load time by 40% (not in the original CV)	by [Add a measurable result, e.g. X%]
November 30, 2024	11/2024 (Europass MM/YYYY)
Current (ongoing degree)	Expected 08/2026
H.S.C, 4.67	Higher Secondary Certificate (HSC), GPA 4.67 / 5.00, EQF level 4
Religion, date of birth	removed — not part of Europass

Placeholders in [brackets] show the user exactly what to fill in, so the CV stays truthful.

Authentication, Security and API



Security

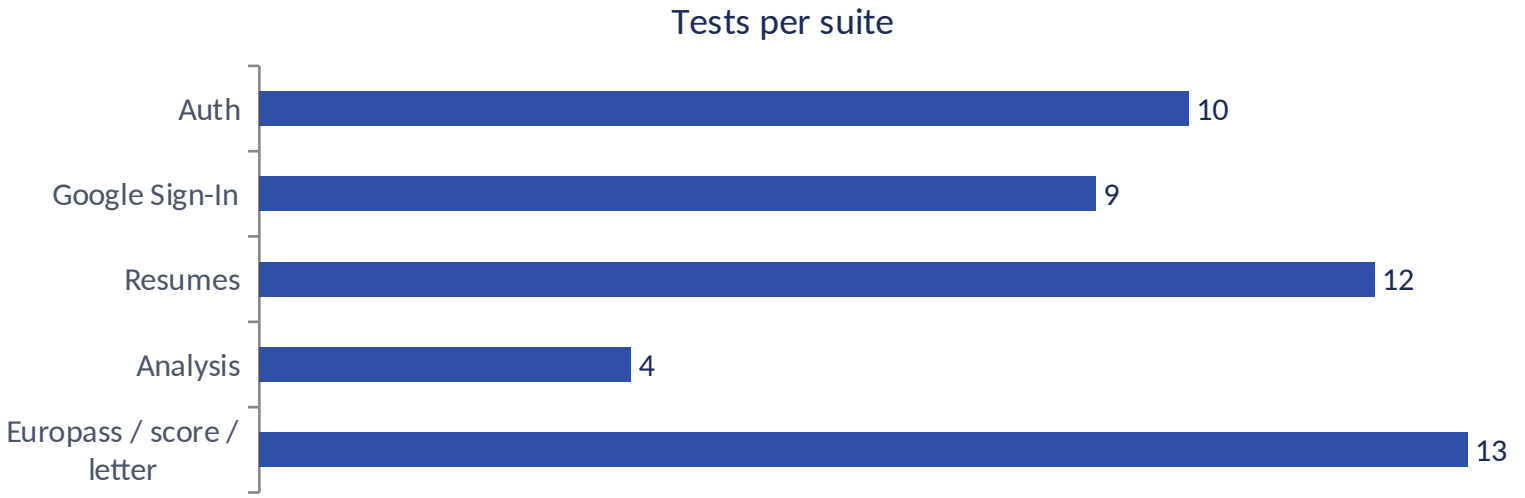
- bcrypt password hashing (8–72 characters)
- Password change logs out other devices
- helmet, CORS allow-list, input validation
- Rate limits: 20 auth / 15 min, 15 AI / hour

API group	Endpoints
Auth (6)	register, login, google, me, password, delete
Resume (6)	upload, list, get, file, image, delete
AI (8)	analyze, improve, CV get/edit, score, cover letter
Health (1)	/api/health

Testing and Continuous Integration

48

automated API tests, all passing in GitHub Actions



Vitest + Supertest

Real HTTP requests against the Express app

In-memory MongoDB

Fresh database on every run

Gemini mocked

Fast and repeatable, no API key

CI on every push

Tests + frontend type check and build

Deployment

Vercel — Frontend

https://ai-resume-analyzer-psi-lime.vercel.app

Render — Backend API

resumind-api-3kpd.onrender.com

MongoDB Atlas — Database + GridFS files

M0 cluster

GitHub Actions — tests before every deploy

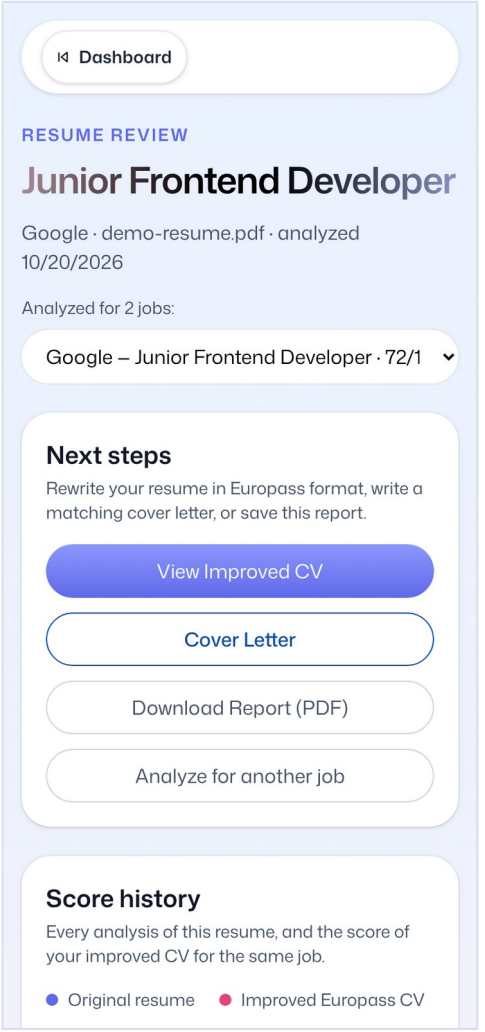
github.com/mahamud894/AI_Resume_Analyzer

Scan to open the live app

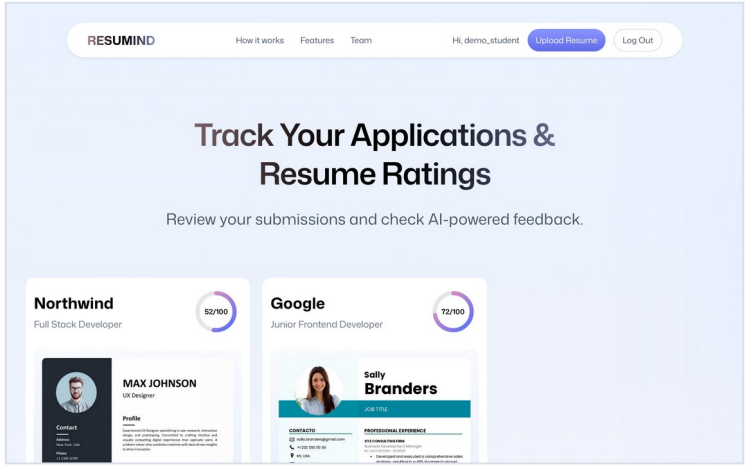
Secrets are kept in .env and the Render / Vercel dashboards — never in Git.

Live Demonstration

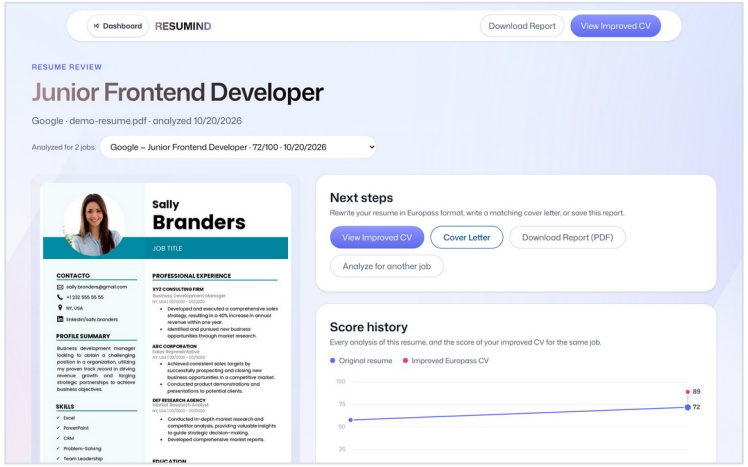
1. Open the live site on laptop and phone
   Mahammad
2. Sign up / Sign in with Google
   Ratul
3. Upload a resume for a real job
   Sakib
4. Read the scores and tips
   Sakib
5. Improve to Europass, edit a line
   Sakib
6. Score the improved CV
   Mahammad
7. Generate and download the cover letter
   Mahammad
8. Dashboard history, CI and deployment
   Himel & Ratul



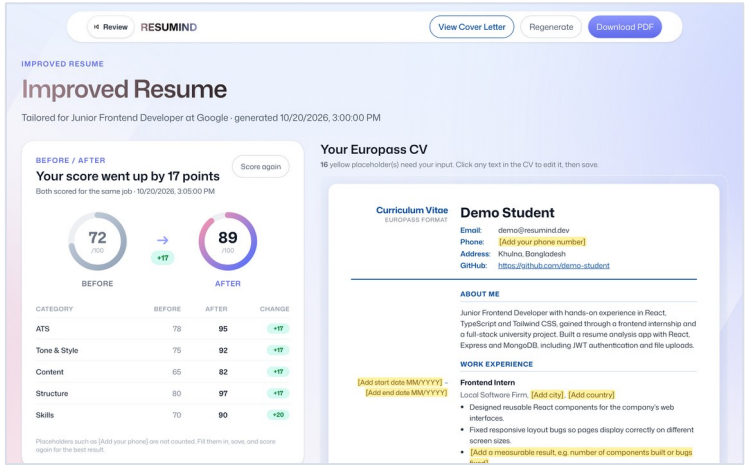
Screens (Backup for the Demo)



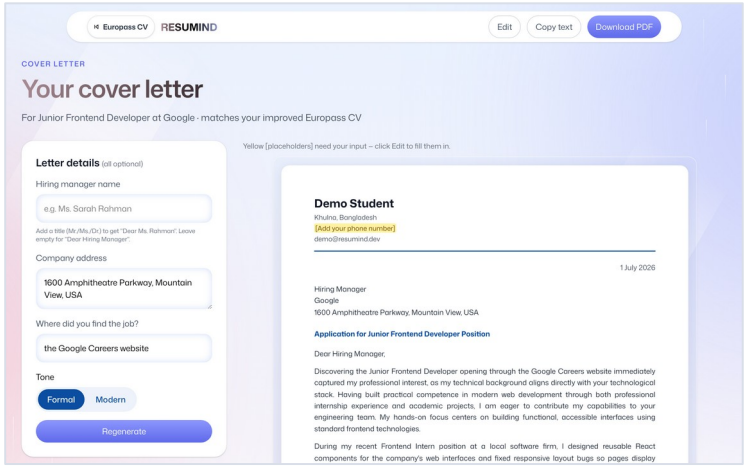
Dashboard



Analysis result



Europass CV



Cover letter

Challenges and Solutions

Challenge	How we solved it
Inconsistent AI output	Gemini JSON mode + strict schema + validation in code
AI invented numbers and dates	Deterministic post-check that inserts placeholders
Gemini "high demand" errors	Automatic retries with back-off and clear messages
Uploaded files lost on redeploy	Moved file storage to MongoDB GridFS
One rate limit for all users behind Render's proxy	TRUST_PROXY so each visitor's real IP is counted
Document-heavy pages on phones	Mobile-first layout and A4 sheets that scale

Future Improvements

OCR for scanned PDFs

Read image-only resumes and DOCX files

Job link import

Paste a LinkedIn or BDJobs link

Interview practice

AI questions from the CV and the job

Bangla support

Interface and feedback in Bangla

Native mobile app

React Native on the same REST API

Recruiter mode

Rank many resumes for one job

Conclusion

- RESUMIND turns a confusing step of job hunting into a guided, explainable process
- All eight objectives of the proposal are achieved
- Own secure backend, database following the ERD and four AI features with safeguards
- 48 automated tests with continuous integration
- Publicly deployed and usable on any device

Live app
ai-resume-analyzer-psi-lime.vercel.app

Source code
github.com/mahamud894/AI_Resume_Analyzer

Thank You

Questions?



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